



Installation and Commissioning Instructions

Automation Marine
mtu NautIQ CoDirect

E532912/01E



A Rolls-Royce
solution

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Table of Contents

Preface

Intended Audience for This Document	v
Associated Documents	v
Book Conventions	vi

Chapter 1: Overview

Introduction	1-2
System Overview	1-2
mtu NautIQ CoDirect Connections	1-3
Installation Overview	1-4
Customer-Provided Components	1-4
Minimum Installation Requirements	1-5
Safety Considerations	1-6
Warnings and Cautions	1-6

Chapter 2: Planning Your Installation

Introduction	2-2
Installation Planning Worksheet	2-2
Locating Installation Procedures	2-5

Chapter 3: Mechanical Installation

Overview of the mtu NautIQ CoDirect Mechanical Installation	3-2
Installing the mtu NautIQ CoDirect Console on Your Vessel	3-3
Installing the Receiver/Antenna for the mtu NautIQ CoDirect Remote Helm	3-5
Mounting Helm Controls	3-6

Chapter 4: Electrical Installation

Introduction	4-2
Connection Types Used in mtu NautIQ CoDirect	4-3
Ports Available on the mtu NautIQ CoDirect Console	4-3
Port Connectors	4-4
Minimum System Connection Requirements	4-5
Connecting Power to the mtu NautIQ CoDirect (Port A10)	4-6
Connecting the Propulsion System to the mtu NautIQ CoDirect	4-7
Connecting the Propulsion System Using CAN Bus (Port A14)	4-7
Connecting the Propulsion System Using an Analog Connection (Ports A3 & A4)	4-10

Connecting the Steering System to the mtu NautIQ CoDirect	4-13
Connecting the Steering System Using CAN Bus (Port A14)	4-13
Connecting the Steering System Using Analog Connections or Digital Relays (Ports A5 & A6).....	4-15
Connecting the Vessel's System Stop to the mtu NautIQ CoDirect (Port A17)	4-18
Connecting the mtu NautIQ CoDirect Remote Helm to the mtu NautIQ CoDirect (Port A15).....	4-19
Connecting the mtu NautIQ CoDirect Ethernet Network Ports (A12)	4-20
Connecting Serial Sensors Using NMEA 0183 (Port A16)	4-20
Connecting Marine Instruments to NMEA 2000 (Port A13)	4-22
Connecting a Bow or Stern Thruster to the mtu NautIQ CoDirect	4-25
Connecting the Bow or Stern Thruster Using the NMEA 2000 Network (Port A13)	4-25
Connecting the Bow or Stern Thruster with Discrete I/O (Port A7)	4-25
Connecting Bucket Control to the mtu NautIQ CoDirect.....	4-26
Connecting Bucket Control Using CAN Bus (Port A14)	4-26
Connecting Bucket Control Using the NMEA 2000 Network (Port A13)	4-26
Connecting Auxiliary I/O	4-27
Connecting Discrete Outputs (Ports A1, A2, A8, A9, and A11).....	4-27
Discrete Inputs (Ports A9 and A11)	4-28
Auxiliary Analog Inputs.....	4-29

Chapter 5: Commissioning the mtu NautIQ CoDirect

Commissioning Your mtu NautIQ CoDirect	5-1
Rolls Royce Power Systems AG mtu NautIQ CoDirect Commissioning Checklist.....	5-2
Dockside	5-2
Underway	5-2

Appendix A: Specifications

Appendix B: Ports on the mtu NautIQ CoDirect Connector Plate

Appendix C: Contact Person/Service Partner

Service	C-1
Local Support.....	C-1
24-h Hotline.....	C-1
Spare Parts Service	C-1

Preface

Welcome to the Rolls Royce Power Systems AG *Installation and Commissioning Instructions mtu NautIQ CoDirect*.

Intended Audience for This Document

The *Installation and Commissioning Instructions mtu NautIQ CoDirect* is intended for users with mechanical, electronic, and networking knowledge. The mtu NautIQ CoDirect must be installed by a qualified marine electronics technician. For electrical installations, the installer must comply with the NMEA 0400 Installation Standard. The installer must also comply with ISO/TR 10134:2020, Small craft — Electrical devices — Established practices for the design, construction and installation of lightning-protection systems, for lightning protection.

The Commissioning procedure that takes place after installation will be performed by Rolls Royce Power Systems AG personnel. They will assist you in performing the sea trials at the end of the commissioning procedure.

Note: Only qualified personnel who are fully informed of the system safety hazards should install or operate this equipment.

Associated Documents

- *Operating Instructions mtu NautIQ CoDirect*

Book Conventions

This guide contains the following information highlights and cross-references:



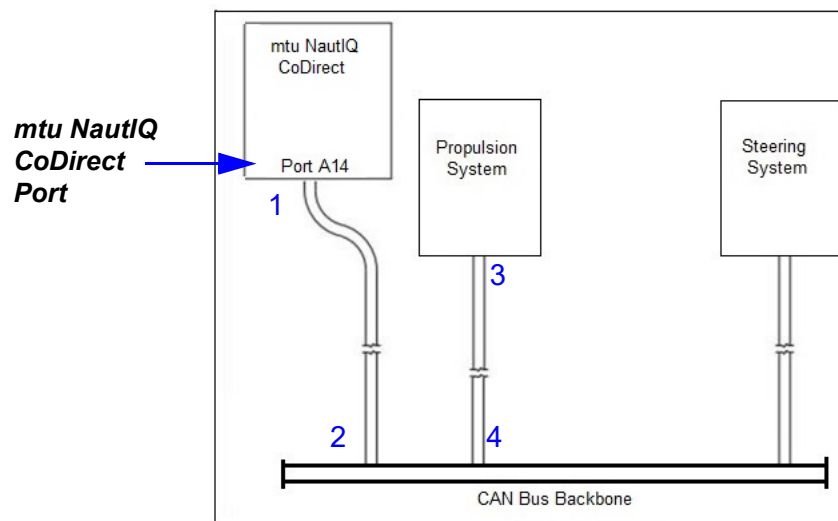
Warning: Warns the user regarding actions that may result in physical damage to the system or personal injury.



Caution: Cautions the user regarding actions that may result in operational issues or data loss.

Note: Identifies important points, helpful hints, special circumstances, or alternative methods.

- Cross references indicate the location of additional information regarding the chosen topic. References may include either headings on specific pages or entire chapters, for example, See “**Introduction**” on page 1-2.
- Buttons, field names, and selections are shown in **bold**.
- Blue boxes, text, and arrows highlight areas of interest on screenshots or images as shown:



Chapter 1: Overview

Topics described in this chapter include:

Introduction	page 1-2
System Overview	page 1-2
mtu NautIQ CoDirect Connections	page 1-3
Installation Overview	page 1-4
Customer-Provided Components	page 1-4
Minimum Installation Requirements	page 1-5
Safety Considerations	page 1-6
Warnings and Cautions	page 1-6

Introduction

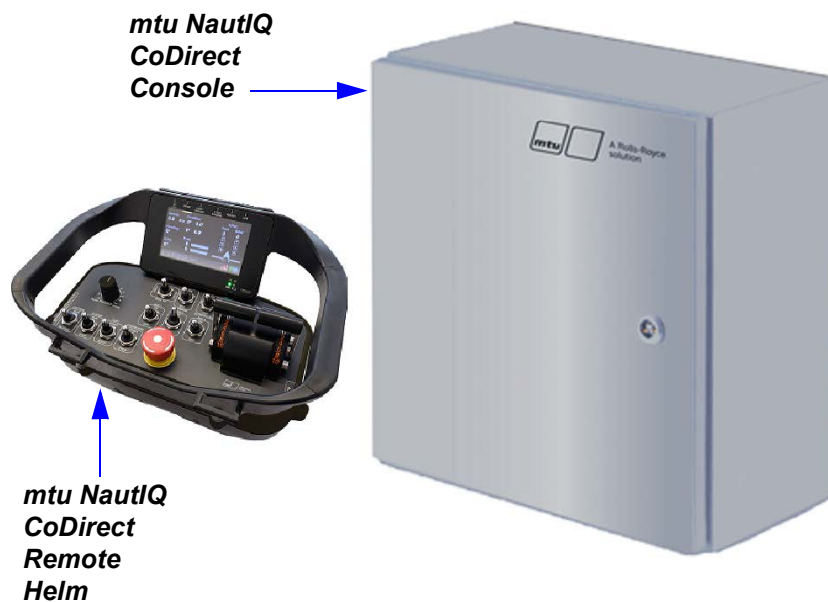
This guide describes how to plan for and install your mtu NautIQ CoDirect system.

System Overview

The mtu NautIQ CoDirect system includes two major components (see **Figure 1-1**):

- mtu NautIQ CoDirect Console
- mtu NautIQ CoDirect Remote Helm Controller (mtu NautIQ CoDirect Remote Helm)

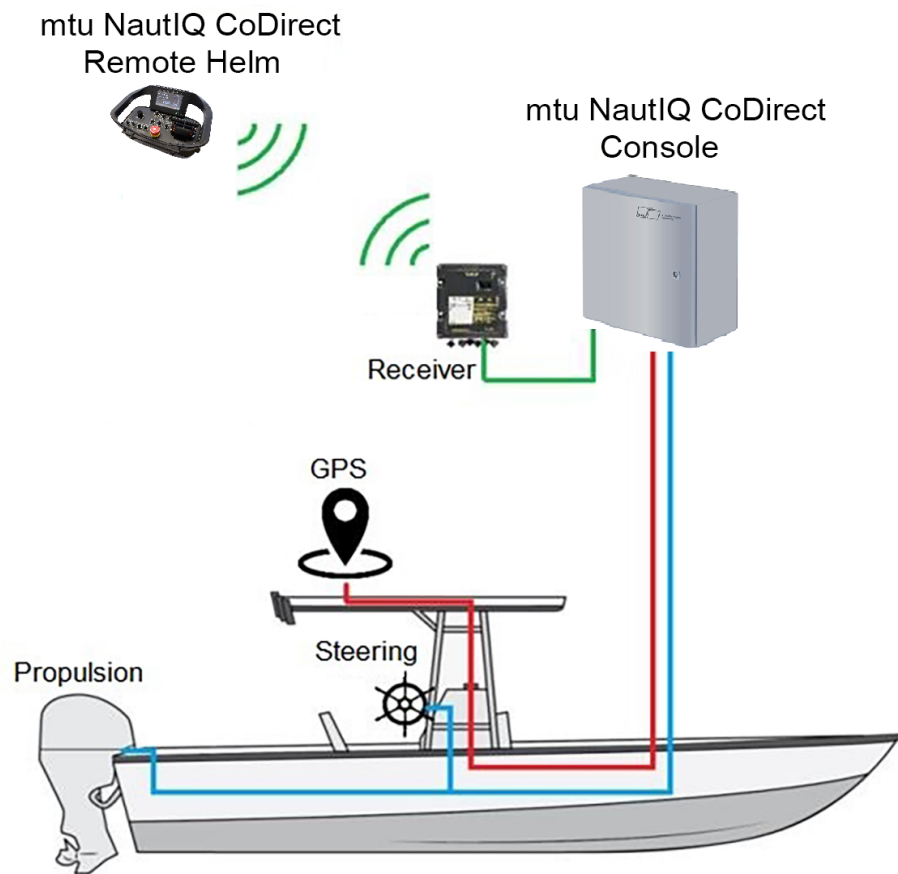
Figure 1-1: mtu NautIQ CoDirect System



mtu NautIQ CoDirect Connections

Figure 1-2 shows an overview of the mtu NautIQ CoDirect connections.

Figure 1-2: mtu NautIQ CoDirect Connections

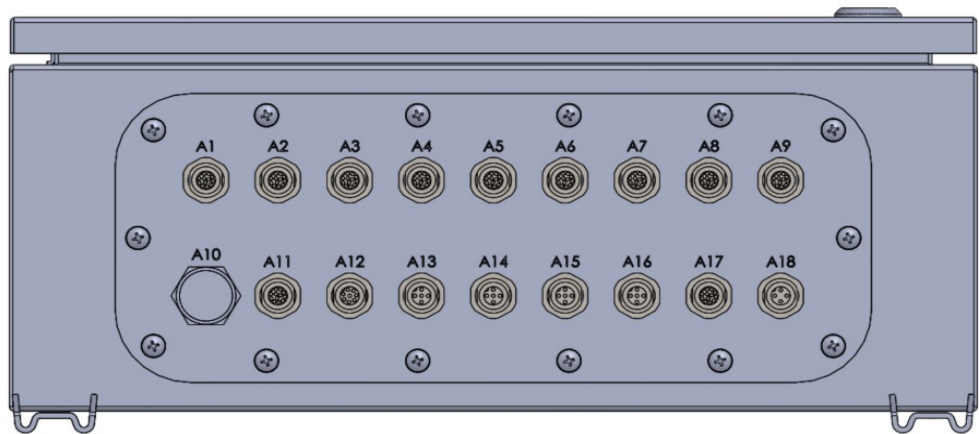


The mtu NautIQ CoDirect Console is mounted within your vessel. (The mtu NautIQ CoDirect Console is designed to be installed in a dry and environmentally protected space within the vessel. If there is no option to install within such a space, the mtu NautIQ CoDirect Console should be installed in a location that protects it from adverse weather conditions, water spray, and direct sunlight, and it should be mounted high enough to avoid any standing water.) The mtu NautIQ CoDirect Remote Helm communicates to the mtu NautIQ CoDirect via a receiver that connects to the Console.

The vessel's steering and propulsion systems and other auxiliary equipment connect to the mtu NautIQ CoDirect Console.

Figure 1-3 shows the connector plate on the bottom of the mtu NautIQ CoDirect Console through which all connections are made.

Figure 1-3: Connection Plate on mtu NautIQ CoDirect Console



Refer to the *mtu NautIQ CoDirect User’s Guide* for a full system overview.

Installation Overview

There are three main parts to the mtu NautIQ CoDirect installation:

- Mechanical Installation
- Electrical Installation

Additionally, after installation, a Commissioning process is performed by a Rolls Royce Power Systems AG engineer or an authorized representative.

Customer-Provided Components

The following components are provided by the customer:

Component	Model	Description
Custom Cables	N/A	Custom cables to connect your vessel components to the mtu NautIQ CoDirect (M12 A-coded connector on one end to flying leads on the other end)